Products of secants and tangents — 3.2

If we want $u = \tan(x)$, then $du = \sec^2(x)\,dx$

and if instead $u = \sec(x)$, then $du = \sec(x)\tan(x)\,dx$

Example: Find $\int \sec^2(x)\tan(x)\,dx$

$u = \sec(x)$

$du = \sec(x)\tan(x)\,dx$

$$= \int u\,du = \frac{u^2}{2} + C = \frac{\sec^2(x)}{2} + C$$

$u = \tan(x)$

$du = \sec^2(x)\,dx$

$$= \int u\,du = \frac{u^2}{2} + C = \frac{\tan^2(x)}{2} + C$$

differ by $\frac{1}{2}$

$\sin^2(x) + \cos^2(x) = 1$ dividing by $\cos^2(x)$ gives

$\tan^2(x) + 1 = \sec^2(x)$, $\tan^2(x) = \sec^2(x) - 1$

Example: Find $\int \tan^4(x) \sec^4(x)\, dx$.

$$= \int \tan^4(x) \left(\tan^2(x)+1\right) \sec^2(x)\, dx$$

$u = \tan(x)$

$du = \sec^2(x)\, dx$

$$= \int u^4 (u^2+1)\, du = \int u^6 + u^4\, du = \frac{u^7}{7} + \frac{u^5}{5} + C$$

$$= \frac{\tan^7(x)}{7} + \frac{\tan^5(x)}{5} + C$$

Example: $\int \tan^5(x) \sec^3(x)\, dx = \int \tan(x) \left(\sec^2(x)-1\right)^2 \sec^3(x)\, dx$

$u = \sec(x)$

$du = \sec(x)\tan(x)\, dx$

$$= \int (u^2-1)^2 u^2\, du = \int (u^4 - 2u^2 + 1)\, u^2\, du$$

$$= \int u^6 - 2u^4 + u^2\, du = \frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3} + C$$

$$= \frac{\sec^7(x)}{7} - \frac{2\sec^5(x)}{5} + \frac{\sec^3(x)}{3} + C$$

Remains to say how to handle an integral w/ an even power of $\tan(x)$ and an odd power of $\sec(x)$.

Magic trick for $\int \sec(x)\,dx$:

$$\int \sec(x)\,dx = \int \sec(x) \cdot \frac{\sec(x) + \tan(x)}{\sec(x) + \tan(x)}\,dx$$

$$= \int \frac{\sec^2(x) + \sec(x)\tan(x)}{\sec(x) + \tan(x)}\,dx \qquad \begin{aligned} u &= \sec(x) + \tan(x) \\ du &= (\sec(x)\tan(x) + \sec^2(x))\,dx \end{aligned}$$

$$= \int \frac{1}{u}\,du = \ln|u| + C = \ln|\sec(x) + \tan(x)| + C$$

$$\boxed{\int \sec(x)\,dx = \ln|\sec(x) + \tan(x)| + C}$$